The Electron Transport Chain

Unit 3B Slides 27 through 36 Honors High School Biology

Slide 27: Stage 4: The Electron Transport Chain

The electron transport chain, or ETC, is the final stage of aerobic cellular respiration. It uses the high-energy electrons collected by NADH and FADH2 to produce most of the ATP made from one glucose molecule.

This stage depends indirectly on oxygen. Oxygen does not supply energy to the chain, but it keeps electrons moving by accepting them at the end.

Slide 28: Summary of Reactions So Far

Before the ETC begins, glycolysis, pyruvate oxidation, and the Krebs cycle have produced only four ATP directly. Their greater contribution is 10 NADH and 2 FADH2, which carry high-energy electrons and hydrogen ions to the inner mitochondrial membrane.

The cell will use the energy in these electrons to establish a proton gradient. That gradient becomes the immediate source of energy for large-scale ATP production.

Slide 29: Structures of the ETC

The ETC is embedded in the cristae, the folds of the inner mitochondrial membrane. These folds increase surface area and provide room for many transport proteins and ATP synthase enzymes.

The protein complexes pass electrons from one carrier to the next and use the released energy to pump hydrogen ions from the matrix into the intermembrane space. ATP synthase provides a controlled route for those ions to return.

Slide 30: Stage 1: Building a Proton Gradient

NADH and FADH2 donate electrons to the ETC and become NAD+ and FAD again. Recycling these carriers matters because glycolysis and the Krebs cycle require the oxidized forms to continue accepting electrons.

As electrons move through the protein complexes, they lose energy. The complexes use that energy to pump H+ ions across the inner membrane. This creates both a concentration difference and an electrical difference, together called an electrochemical gradient.

Slide 31: Stage 2: Oxidative Phosphorylation

Hydrogen ions then diffuse down their gradient through ATP synthase. Their movement turns the enzyme much like water turns a turbine, allowing ATP synthase to add phosphate to ADP. This coupling of ion movement to ATP production is called chemiosmosis.

At the end of the chain, oxygen accepts low-energy electrons and combines with hydrogen ions to form water. Because oxygen accepts electrons last, it is the final electron acceptor.

Slide 32: Following the ETC Diagram

Use the diagram to trace the sequence from electron donation to ATP production. Electrons move through the membrane proteins, pumping H+ into the intermembrane space, and H+ returns through ATP synthase. Oxygen then accepts the electrons and helps form water.

Keep the two movements separate: electrons travel along the chain, while hydrogen ions move across the membrane and then through ATP synthase. The proton gradient, rather than the electrons directly, powers ATP synthase.

Slide 33: ETC Video

As you watch, identify where NADH and FADH2 donate electrons, how the proton gradient develops, and where ATP synthase produces ATP. Also note the point at which oxygen accepts electrons and water forms.

Slide 34: What Happens Without Oxygen?

Without oxygen, the ETC cannot pass electrons along. The chain becomes backed up, NADH and FADH2 cannot unload their electrons, and the cell cannot regenerate enough NAD+ and FAD for the earlier stages of respiration. Oxidative phosphorylation stops, so ATP production falls sharply.

The cell then relies on fermentation. Fermentation does not replace the ATP output of the ETC. Its primary purpose is to regenerate NAD+ so glycolysis can continue producing a small amount of ATP.

Slide 35: Summary of Cellular Respiration

This diagram connects the entire pathway. Glucose provides the original carbon and high-energy electrons. Carbon dioxide is released during pyruvate oxidation and the Krebs cycle, while oxygen enters the ETC and becomes part of water.

ATP comes from two mechanisms. Glycolysis and the Krebs cycle make a small amount through substrate-level phosphorylation, while the ETC makes most ATP through oxidative phosphorylation. Heat is also released as energy transfers occur, which helps explain why cellular respiration is not completely efficient.

Slide 36: Summarizing the Outputs

Have students use the notes diagram to account for the major products. Per glucose, glycolysis produces 2 ATP and 2 NADH. Pyruvate oxidation produces two NADH and two carbon dioxide. The Krebs cycle produces 2 ATP, 6 NADH, 2 FADH2, and 4 carbon dioxide.

The ETC uses the ten NADH and two FADH2 to generate most of the remaining ATP, commonly summarized in this course as approximately 32 to 34 ATP from the ETC and about 36 to 38 ATP total. Emphasize that exact totals can vary because cells differ in how efficiently they transfer energy into the mitochondria.